

B3LYP/def2-SVP S1 excitation energy / oscillator strength

Molecule	Solver	PySCF	GradTDDFT
H2O	TDA	7.6 / 1.8e-02	7.6 / 1.8e-02
H2O	TDDFT	7.6 / 1.8e-02	7.6 / 1.8e-02
CO	TDA	8.7 / 1.1e-01	8.7 / 1.1e-01
CO	TDDFT	8.5 / 8.2e-02	8.5 / 8.2e-02
N2	TDA	9.5 / 4.5e-31	9.5 / 3.4e-31
N2	TDDFT	9.4 / 4.7e-18	9.4 / 7.2e-25
C2H4	TDA	8.3 / 5.4e-29	8.3 / 6.0e-29
C2H4	TDDFT	8.1 / 3.7e-01	8.1 / 3.7e-01
CH2O	TDA	4.0 / 1.9e-29	4.0 / 1.8e-29
CH2O	TDDFT	4.0 / 2.2e-25	4.0 / 5.4e-29
C6H6	TDA	5.5 / 3.7e-09	5.5 / 3.7e-09
C6H6	TDDFT	5.5 / 4.9e-09	5.5 / 4.9e-09

Each cell reports Omega1 (eV) / f1. Values are rounded for manuscript display.